The Digi-Comp II: counting, adding, multiplying, subtracting, dividing

West bay 3: balls that compute in binary · 31 October 2026

`README.pdf` beside this file is a typeset copy of it. `MANUAL.md` (and `MANUAL.pdf`) is an illustrated manual: how each operation works, ball by ball, and which parts each one needs.

Babbage's Hall, west bay 3. The Digi-Comp II, designed by John Godfrey in 1965 and made by E.S.R. Inc. in the late 1960s, was a computer of marbles: a sloping two-level board of flip-flops and switches that balls rolled down one at a time. Set up with its switches, it counted, cleared, added, subtracted, multiplied and divided binary numbers. Its manual names four registers: the **accumulator** (7 flip-flops), the **memory** (4 bits), the **MQ** (3 flip-flops: multiplier and quotient) and the **distributor** (3 flip-flops), and the switches MULTIPLY, CLEAR, COUNT, COMPLEMENT, OVERFLOW and AUTO/MANUAL.

It is being built here in steps. The first (October 2026) was a counter of three flip-flops; the second (2a) the accumulator, with START, CLEAR and COUNT; then (2b) **the memory, the MQ, the distributor and MULTIPLY**, to add and multiply; then (2c) **COMPLEMENT**, to subtract and divide. This, then (2d) **AUTO**: a ball reaching the bottom lets the next one go, so a whole multiply runs by itself from one press of START. This, step 2e, adds **OVERFLOW**'s switch, so that counting out and dividing run by themselves too.

What it does

Two levels, as the original: the balls compute on the upper plate, and a ball that is done drops through a hole to the lower level, which carries it under everything else down to the **collector** at the bottom. The upper plate is drawn see-through, so the lower level shows.

The flip-flop is a rocker on a pin: a roof with a cup at each end and a tail onto two stop pegs, with a small steel ball pressed in under the apex. Its centre of mass is a little up the slope from the pin, so it falls over to one side or the other and stays there: a memory of one bit. Apex to the right reads **1**, to the left **0** (printed either side of each apex). A ball rolling straight down meets the roof on its raised side, rolls into that cup, and its weight takes the rocker over; at the end of the swing the cup tips it out. A ball that finds a flip-flop at 0 sets it to 1 and leaves to the right; one that finds it at 1 sets it to 0 and leaves to the left. Nothing else turns the rockers: their shape and weight do it. All thirteen are the same part; where each side leads makes the register.

The accumulator, A is seven of them down a staircase, A1 (the 1s) at the top right to A7 (the 64s):

- A ball that finds a flip-flop at **0** sets it to 1 and leaves to the right, through that flip-flop's **done hole**, to the lower level.
- One that finds it at **1** sets it to 0 and leaves to the left, down to the next flip-flop: the **carry**.

So a ball let into A1 adds one, carrying as far as it must; a ball let into A3 adds four. A carry out of A7 drops through the **overflow** hole.

CLEAR. Under each of A's rockers, on the lower level, is its **foot**: an arm up the slope from the pin, joined to the rocker by a stem through the plate, ending in a wedge. When the rocker reads 1 the wedge leans into the **clear lane** (green), which runs down the lower level past every foot. A ball sent down the

clear lane meets each wedge in its way and pushes it aside, which turns that rocker to 0. So one ball clears the whole accumulator. (The other six flip-flops have their feet too, being the same part, but no lane beside them.)

The switches are flaps, set by hand. Each sits at the tip of the divider between two outlets, its free end against one wall of the inlet, closing that outlet: a ball only presses it harder against the wall and leaves by the other. A ball from START meets **MULTIPLY** first. Off, it runs down to **CLEAR**: on, it drops into the clear lane; off, it goes on to **COUNT**: on, it runs into A1 and adds one; off, it drops to the lower level and counts nothing.

The memory, M is four switches, M1 (1s) to M4 (8s), one beside each of A1 to A4. Set to 1, a ball that reaches it runs into that flip-flop of the accumulator (into its funnel, by the funnel's long left wall): M3 at 1 adds four. Set to 0, the ball drops through a hole beside it, and rolls down the lower level to the **return lane** at the board's left edge, and on to the collector.

MULTIPLY on sends the balls to **the distributor, D**: D1 sends them to D2 and D3 by turns, and each of those by turns to its two sides, so four balls go four ways, and the fifth begins again. Set to start (D1 1, D2 1, D3 0), the first ball of every four goes from D2 into **the MQ**, the second to M1, the third to M3, the fourth to M2.

The MQ (multiplier-quotient) is three flip-flops down a staircase like the accumulator's, MQ1 (1s) to MQ3 (4s), but read the other way: **1 with the apex to the left** (it says so beside each). So a ball through it takes one away: the borrow runs on to the left. A ball that leaves the MQ on the right (done) runs on, through the funnels under MQ2 and MQ3, to M4. A borrow out of MQ3, when the MQ was 0, drops through **STOP**.

So with MULTIPLY on, every four balls add the memory to the accumulator, once for each count of the MQ, and when the MQ is used up the next ball drops through STOP and the hand stops turning: the MQ times the memory, added to what was in the accumulator, in 4 x MQ + 1 balls. **ADD** is the same with the MQ at 1 (five balls), as in E.S.R.'s manual.

COMPLEMENT. Below each of A1 to A6's done holes a **done lane** runs down and across into the next flip-flop's funnel. As a rule the hole is open and the ball drops through it. COMPLEMENT is six **shutters**, one over each of those holes, on one bar high over the walls (its handle beside A1, `X`): slid left (on), they cover the holes, and a ball that sets a flip-flop to 1 rolls over its hole and down its done lane into the next, as a carry would. So it goes through every flip-flop from where it came in to A7 and turns each over. With COUNT on, one ball makes A 127 - A; COMPLEMENT off, one more adds 1: 128 - A, the two's complement. Add that to a number and the sum goes over 128 (the carry drops through OVERFLOW and is lost): it has subtracted. **Dividing** is adding the divisor to the dividend's complement, cycle by cycle, while the MQ (set to 7) counts the cycles, until A goes over: in MANUAL the hand watches the OVERFLOW hole and stops at the end of that cycle; in AUTO, with OVERFLOW set to HALT, the machine stops itself there (below). The MQ read the other way gives the cycles; if A reads 0 that is the quotient, otherwise the quotient is one less and the remainder is M - A (`MANUAL.md`, section 6). E.S.R.'s machine does this with six more flip-flops (T1 to T6) and two to sequence them (CF1, CF2) in four balls; this one does it with the shutters in two balls, the hand moving COMPLEMENT between them.

START is the feeder: a wheel with four notches under the **input** channel of waiting balls, each quarter turn carrying one ball round from the top and letting it out at the bottom. A wound **weight** turns it, as fast as a **fly** lets it, and an **anchor** holds it, as a clock's escapement holds its wheels: two fingers on the anchor catch the pins of an **escape star** on the wheel. Push the end of the anchor's long arm (the START lever) and it lets the wheel round one quarter turn, one ball; let go and a counterweight (a 1/2 inch steel ball on its other arm) takes it home.

AUTO. At the bottom, the collector's wall turns every ball into a **trip channel** down the right-hand side, and on to the **store** below it, where they wait for the hand. In the channel is the **trip lever**: a short arm on an axle across the channel, trailing down the slope, resting just above a ball's middle. A ball rolls under the axle and under the arm, lifting it 4 mm. The lever's axle comes out through the rim, beside the board, where an arm hangs down from it: lifting the lever swings that arm's foot down the slope. A **thread** tied there runs straight up beside the board's right edge, outside everything, to an **eye** on a post at the edge level with the START lever, turns, and runs across to the START lever, 19 mm from the anchor's pin: it pulls the lever to the right, as the hand does, and the anchor trips. It only ever pulls. **AUTO**: the thread hooked into the START lever, and one press of START runs a whole multiply, ball after ball. The ball through **STOP** drops into a pocket under the plate and never reaches the lever, so the machine stops by itself. **MANUAL**: the thread unhooked (on a peg beside the lever); the trip lever still lifts for every ball, nothing follows, and every ball is a press of START.

OVERFLOW is a shutter over the overflow hole, the way out of A7 to the left, sliding up and down the slope in a groove; its knob stands up the slope and to the left, out of every ball's way. **RUN** (closed, the knob down the slope): a carry out of A7 rolls over the hole, down a lane to the plate's end, and on to the collector and the trip lever, as every ball does. **HALT** (open, the knob pushed up the slope): it drops through into a pocket on the lower level and reaches no trip lever, so in AUTO the machine stops there, as at STOP. That is E.S.R.'s way of counting out a number of balls (A set to its complement, COUNT on, OVERFLOW HALT, START once) and of dividing (the machine stops at the overflow; the MQ read the other way then gives the cycles begun).

Using it

Key	
`Space`	START: push the START lever, one ball (in AUTO: and on by itself, until a ball drops through STOP)
`U`	AUTO and MANUAL: the thread hooked into the START lever, or off
`O`	OVERFLOW: RUN and HALT (its shutter closed, or open)
`M`, `C`, `N`	MULTIPLY, CLEAR, COUNT on and off (by hand)
`1` to `7`	set that flip-flop of the accumulator over by hand (`1` the 1s ... `7` the 64s), as you set the original's
`H`, `J`, `K`, `L`	the memory's 1s, 2s, 4s, 8s (M1 to M4) on and off
`8`, `9`, `0`	set the MQ's 1s, 2s, 4s over by hand
`D`	set the distributor to start a cycle (D1 1, D2 1, D3 0)
`X`	COMPLEMENT on and off (the shutters over the done holes)
`G`	run the program the **program** slider chooses (`G` again stops it)
`B`	the camera to the program board, close enough to read it; again, back
`T`	the tour: every program in the library, in turn (it starts with it)
`R`	set it back by hand: the balls to the top, everything to 0, COUNT on, the others off, OVERFLOW RUN, MANUAL
`Z`	slow motion (1/4 speed) on and off

To multiply: set the accumulator to 0 (or CLEAR it), the memory to one number and the MQ to the other, `D`, MULTIPLY on, and `Space` until a ball drops through STOP; or `U` (AUTO) and one `Space`. To add, the same with the MQ at 1 and the accumulator holding the first number. To subtract, complement the number to take away (COUNT on, MULTIPLY off: `X`, `Space`, `X`, `Space`), then add the other.

Where to look.

- **The rockers**, against the 0 and 1 printed either side of each apex (the MQ's the other way round), and the 1, 2, 4 ... 64 beside A's: the machine's own reading. Read A bottom (64s) to top (1s) for the number written the usual way.
- **The switches'** flaps, against the on and off (the memory's 1 and 0) printed at their outlets.
- **The lower level** through the plate: done balls rolling down, a clear ball running down the green lane past the feet, and the memory's dropped balls going to the return lane on the left.
- **The sign** on the table: what the accumulator, the memory and the MQ read, in binary and as numbers, and how many balls have been let in. Under them, **THIS BALL**: before each ball, what it will do ("D1, D3, M1: adds 1") and what the accumulator was and will be. At the same time a **yellow dashed line** on the board shows the way it will go, through the switches and flip-flops, carries and all; then the hand lets it go, and you can watch it do just that (`Z` for slow motion). The sign and the line are the exhibit's, not part of the machine: the script works them out from the rockers and switches, as you would.

Rest the mouse on a rocker, a foot, a flap, the feeder or a wall to see what it is and what it is doing.

The tour (`T`; it starts with the exhibit) runs the program library (below), one program after another: count to 5; clear; **ADD** 37 + 11 (five balls, the last to STOP: 48); **MULTIPLY** 13 x 3 (thirteen balls: 39); 13 x 5 in **AUTO** (one START, twenty-one balls by themselves: 65); the **COMPLEMENT** of 25 (102, then 103); **SUBTRACT** 12 - 4 (4 complemented to 124, then 12 added: 8, over 128); **DIVIDE** 13 / 5 ball by ball (quotient 2, remainder 3), 25 / 5 with loops, and 25 / 5 in AUTO, stopped by OVERFLOW; and counting out nine balls, by hand and in AUTO. Each program's listing is in the Shortcuts pane, the line being done in the status bar. After each ball it checks everything the ball should have changed (the accumulator, the MQ, the distributor, STOP, OVERFLOW) and that the ball reached the collector, and each program's `CHECK` lines, and prints `RESULT digicomp2 tour: PASS`. (In AUTO each ball is checked as it lifts the trip lever, the next already let go.) It takes about 1350 simulated seconds (81 000 frames), and about twenty minutes of computing here, headless. Twelve balls are not enough for the multiply: when the input runs low the hand takes balls from the collector and puts them back at its top, as you would.

Programs. The procedures can be written down and run: text files in `programs/`, one step a line (the keys you would press, and `SET`, `UNTIL`, `OVERFLOW`, `AUTO`, `MANUAL`, `RUN` and `CHECK`), listed in `programs/LIBRARY`. Choose one with the **program** slider and press `G`: the hand does it line by line, every ball shown before it goes and checked after. The program is shown on the **program board**, on a stand behind the machine's top end, the line being done lit (`B` brings the camera close enough to read it, and back); it is also in the Shortcuts pane, and the line in the status bar. The library has count, clear, 37 + 11, 13 x 3, 13 x 5 in AUTO, the complement of 25, 12 - 4, 13 / 5 ball by ball, 25 / 5 with loops and in AUTO, and counting out nine balls by hand and in AUTO; `MANUAL.md`, section 9, has the language.

Sliders: `friction` (0.2 to 0.6 checked), `program`, `speed`, `steps`.

Running it

Open `digicomp2.lua` in bpp, or run it inside the hall (`HALL_START=digi`). Environment, for testing: `DC_TOUR=0` (no tour), `DC_FRICTION`, `DC_PF` (the pins' friction, as a torque in kg cm^2/s^2: 0.4), `DC_TH` (the board's slope, 15 degrees), `DC_ROLL` (the table tilted across, degrees), `DC_HZ` (steps a second, 2400), `DC_WEIGHT` (the weight's torque on the wheel, 30 kg cm^2/s^2), `DC_FLY` (the fly's top speed, 3 radians a second), `DC_PUSH` (the hand's push on the START lever, as a torque on the anchor's pin, 300), `DC_PAUSE` (seconds between a reading and the next ball, 1), `DC_SHOW` (seconds the next ball's way is shown before it is let go, 1.5), `DC_STEP` (begin the tour at that program of the library), `DC_PROGRAM=name` (run only that program; `all`: every one in turn, as the tour, then `RESULT digicomp2 programs: PASS`), `DC_LOG=1` (every rocker's lean at each reading), `DC_LOG=2` (also every moving ball four times a second), `DC_LOG=3` (where

every ball is at each reading), `DC_CONTACTS=frame` (every contact of the balls, rockers, flaps and wheel at that frame), `DC_REST=0` (never rest), `DC_EXPORT=1` (rewrite `parts/geometry.scad`).

`tests/toggle.lua` is one flip-flop alone, with its foot, its done hole and the clear lane; `tests/switch.lua` is one switch alone; `tests/junction.lua` is the piece of the board where the memory meets the accumulator (A1, A2 and M2: the carry and the memory's balls into the same funnel); `tests/complement.lua` is A1 to A3 with their done lanes and shutters, each ball checked with the shutters open and closed; `tests/escapement.lua` is the feeder wheel driven by a weight and held by the anchor, alone (the trip lever and thread are not in it). Each prints `RESULT` lines; their headers list their settings, and `FINDINGS.md` has what they showed.

Printing and building

`./make-stls.sh` (needs OpenSCAD, and bpp to refresh `geometry.scad`) writes every part into `stl/`, each turned to lie on the bed. Every shape a ball, a rocker or a flap touches comes from `digicomp2-geometry.lua`, the same numbers the simulation uses. The largest pieces are the tiles, 214 x 189 mm (on a 220 mm bed, or turned on a 250 x 210 one).

Part	Print	Size (mm)	Notes
`upper-i-j.stl`	1 each (18)	214 × 176 x 16	the upper plate in tiles, two across and nine down: its 4 mm face with the holes and the shutters' grooves (2 mm deep), the walls (12 mm high), the stop pegs, the bosses under the flaps and the wheel, the lettering cut 0.6 mm deep (fill it with paint, or change filament at that layer)
`lower-i-j.stl`	1 each (18)	214 × 189 x 17 (the bottom two 32)	the lower level in tiles: a 2 mm floor with its walls (15 mm), the clear lane, the return lane, the collector's wall, the trip channel and the store's wall (30 mm), STOP's pocket, the posts that hold the plate up, and the bosses the feet stand on
`rocker.stl`	13	44 × 19 × 9	a flip-flop's top. 100% infill
`foot.stl`	13	12 × 24 × 22	its stem and foot: the stem glues up into the rocker's hub, through the plate's square hole. 100% infill
`flap.stl`	7	3 × 19 × 9	a switch's flap
`wheel.stl`	1	39 × 39 × 9	the feeder's wheel
`star.stl`	1	40 × 40 × 8	the escape star, glued on the wheel's hub (its stem is where the weight's drum and the fly will go)
`anchor.stl`	1	55 × 91 × 8	the anchor: its two fingers, the START lever, the counterweight's arm with its cup, and the hole in the START lever the thread hooks into
`trip-lever.stl`	1	14 × 10 × 10	AUTO: the trip lever
`trip-down-arm.stl`	1	3 × 17 × 4	AUTO: the arm hanging from the trip lever's axle, beside the board; the thread is tied through the hole at its foot
`trip-eye.stl`	1	8 × 8 × 50	AUTO: the thread's eye, a post glued to the board's right edge, level with the START lever (its 3 mm post may be stiffer as a steel rod)
`shutter.stl`	6	28 × 22 × 2	COMPLEMENT: a done hole's shutter, sliding in its groove in the plate; a pin pressed into its hole
`ovf-shutter.stl`	1	38 × 33 × 20	OVERFLOW: the shutter over the overflow hole, sliding in its groove in the plate, its knob on its arm
`comp-knob.stl`	1	10 × 10 × 12	COMPLEMENT's handle, glued on the top end of its bar
`foot-back-1.stl`, `foot-back-2.stl`	2 each	30 × 228 x 30, 30 × 232 x 30	the stand's tall back feet, each in two halves glued end to end (20% infill is plenty; on a 220 mm bed, lay them across its diagonal)
`foot-front.stl`	2	30 × 21 × 30	its low front feet

Also needed

- **12 steel balls, 1/2 inch (12.7 mm)**, 8.35 g each (and a few spare; with more, a long multiply needs fewer trips to the collector).
- **13 steel balls, 1/4 inch (6.35 mm)**, 1.04 g: the rockers' weights, pressed into the pocket under each apex, flush.
- **3 mm steel dowel pins**: thirteen 32 mm long for the flip-flops (pressed into the lower tiles' bosses, up through foot, stem and rocker), seven 16 mm for the flaps and one 20 mm for the wheel (pressed into the upper plate).
- For COMPLEMENT: six more 3 mm pins, 21 mm long (one in each shutter), and the **bar**: a strip of aluminium (or hardwood) 5 x 3 mm, 792 mm long, drilled 3.4 mm for the pins, the first 30 mm from its top end and the others every 151.6 mm after it (the pins' places are `SH_PINS` in `parts/geometry.scad`).
- A **lower board** of 6 mm plywood, 424 x 1691 mm, the lower tiles glued on it edge to edge, and the feet glued under it at its corners.
- For START and AUTO: one more 3 mm pin, 20 mm (the anchor's); a 1/2 inch steel ball, glued in the anchor's cup (its counterweight); a **2 mm steel wire**, 31 mm, the trip lever's axle, glued into the lever and the hanging arm and turning in the holes in the channel's wall and the rim; a strip of **PTFE tape** on the lever's underside, where the balls lift it; and the **thread**: a fine braided line that doesn't stretch (0.2 mm Dyneema fishing line), about 1.7 m: 148 cm up beside the board and 17 cm across to the START lever, a small wire hook on its end.
- Glue; a little dry PTFE or graphite on the pins.

Assembly. Glue the lower tiles onto the plywood, edge to edge, the lanes' walls lining up across the joins. Press the flip-flops' pins into their bosses. Glue the upper tiles together at their edges into one plate (a strip of thin card or tape under each join helps). Put each foot on its pin, wedge up the slope; lay the plate on its posts, the stems coming up through the square holes; press the steel balls into the rockers and glue each rocker onto its stem, roof up the slope. Press the flaps' and wheel's pins into the plate, fit them, glue the star on the wheel and fit the anchor on its pin, its counterweight glued in. Fit the trip lever in its channel on its wire axle and glue the hanging arm on the wire's end beside the board. Glue the eye's post to the board's right edge, tie the thread to the hanging arm, lead it up through the eye, and cut it so its hook just reaches the hole in the START lever with everything at rest: hooked in for AUTO, on the peg beside it for MANUAL. Put the shutters in their grooves, pins up, slip the bar over the pins and glue its knob on. Glue the tall feet under the back corners and the low ones under the front, so the board slopes at 15 degrees toward you, and **level it across** (within 1 degree; see `FINDINGS.md`). Set every flip-flop to 0 (apex left; the MQ's read 0 with the apex right), COUNT on and the other switches off, put the balls in the input channel, and push the START lever once for each ball (or, in AUTO, once).

The first push after loading fills the wheel: balls come out from the second. (In the simulation, the hand loads the wheel's two notches as it puts the balls back, so the first lets one out.)

What is not as built

- **The weight and the fly** are not parts yet: in the simulation they are a steady torque on the wheel's pin (`DC_WEIGHT`) and a top speed (`DC_FLY`). A real one needs a drum on the star's stem, a cord and a weight to wind up, and a small fan geared to the wheel.
- **The thread** is simulated as a line that can't stretch, that pulls and never pushes, and that runs through its eye without friction; it weighs nothing and touches nothing. A real one stretches a little and rubs in the eye.

- **The hand.** It pushes the START lever (a push of 0.3 s); sets the switches and flip-flops, puts the balls back from the store and the STOP pocket, by placing them, and in MANUAL watches STOP: it pushes no more once a ball has dropped through. In AUTO the machine stops itself; the hand keeps the input filled from the store as it runs. Dividing in MANUAL, it watches OVERFLOW too, and stops at the end of that cycle. It moves COMPLEMENT between the two balls of a complement, which E.S.R.'s machine does itself with CF1 and CF2.
- **AUTO** is this machine's own: E.S.R.'s ball at the bottom rolls under the START lever itself; here the START lever is at the top, and the trip lever at the bottom works it through the thread. E.S.R.'s OVERFLOW switch steers the ball at the overflow; here it is a shutter over the hole.
- **The layout** is this machine's own, not E.S.R.'s: the registers do what the manual says, but where each part sits on the board was worked out here. The distributor is a tree of three flip-flops, and after the MQ and M4 it serves M1, M3, M2 (the manual's order is M2, M3, M1; the sum is the same). The complement is done with shutters and done lanes, not E.S.R.'s mode flip-flops; it made the board longer (the accumulator's steps are 147.5 mm apart, for the lanes between them).
- **The pins** are hinges with a constant friction torque, 0.4 kg cm^2/s^2 (`DC_PF`): PLA on PLA at a boss, worked out in `FINDINGS.md`.
- **The lettering** is drawn on the plate's face, not cut into it; the plate is drawn see-through.

See `FINDINGS.md` for how the numbers were arrived at.